

Ahmed Akram Amer

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Shebin el-Kom, Menoufia, Egypt

Electronics Engineering Graduate | 5+ Production AI Systems | LLM Applications & Multi-Agent Systems

Education

Menoufia University

Sep. 2021 – Oct. 2026

Bachelor of Electronic Engineering — Computer Science & Engineering Department | GPA: 3.71 / 4.0 Shebin el-Kom, Egypt

Work Experience

AI & ML Engineer Intern · Flyrank.AI

Jul. 1 – Aug. 26, 2026

Internship — AI and Machine Learning (8 Weeks)

- Learning ML fundamentals both in theory and in practice.

AI Engineering Intern · Tips Hindawi

Jun. 18 – Jul. 18, 2026

Internship — AI Engineering, Egypt (1 Month)

- Learned HuggingFace Transformers architecture (Attention Is All You Need), Groq, Streamlit, LangChain, and RAG.
- Built LLM applications including an HR Q&A; Assistant Chatbot and a Custom-PDF RAG application.

MLOps Intern · SAiR

Internship — MLOps Fundamentals and Best Practices

- Learning the fundamentals of ML, the ML lifecycle, MLOps in theory, and best practices.
- Hands-on tools: FastAPI, Docker, Kubernetes, Contabo, Feast, Prefect, MLflow, AWS.

AI & Data Science Intern · DEPI | EYouth

Jun. 2024 – Dec. 2024

Internship — AI & Machine Learning Egypt (Remote)

- Engineered supervised ML models for predictive analytics, achieving 90% accuracy on structured datasets.
- Reduced manual reporting time by 50% through automated ML pipelines deployed via Flask and Docker.
- Conducted EDA and feature engineering with Pandas and Seaborn across 10+ high-dimensional datasets.

Projects

QueryGuard AI — Enterprise Text-to-SQL Platform

Python, Gemini 2.5 Flash, Streamlit, SQLite3, RBAC

- Built a GenAI platform translating English/Arabic questions into optimized SQL queries for non-technical users.
- Implemented Dual-Layer RBAC: DROP, DELETE, TRUNCATE blocked at both UI and LLM prompt levels.
- Modular architecture supports one-line migration from SQLite3 to PostgreSQL or Amazon RDS.

ReActify-PDF — Chat with PDF RAG Project

Python, FastAPI, LangChain, Qdrant, SQLite, Gemini API

- Engineered a bilingual (English/Arabic) ReAct agent for PDF Q&A; with source-attributed answers.
- Built lock-free SQLite-backed Qdrant ingestion pipeline achieving document indexing in seconds.

Multi-Agent Research & Writing System

Python, LangGraph, CrewAI, FastAPI, n8n, Groq LLaMA 3.3

- Built a multi-agent pipeline cutting research time by ~80%, generating structured reports in under 2 minutes.
- Integrated Groq LLaMA 3.3 70B achieving GPT-4-comparable output; resolved 6+ integration challenges.

AI-Powered Procurement System — Multi-Agent

Python, CrewAI, LangChain, Groq LLaMA 3.3, Tavily, AgentOps

- Designed a 4-agent CrewAI pipeline automating procurement workflow, cutting manual research by 70%.
- Containerized with Docker; agents compare prices across Amazon Egypt, Jumia, and Noon in real-time.

Multilingual RAG Chatbot — Customer Support

Python, LangChain, Chroma, Groq, HuggingFace, Gradio

- Built production RAG chatbot handling Arabic/English queries with MMR retrieval (k=6).
- Implemented hallucination control via strict system prompting — agent escalates to human support when needed.

Deep Learning — Multi-Domain | ML Lifecycle Factory

PyTorch, TensorFlow, Scikit-learn, Flask, MLflow, OpenCV

- End-to-end PyTorch pipelines: image classification (CIFAR-10, MNIST), audio classification, breast cancer detection.
- Architected ML Forge: web platform automating full ML lifecycle, reducing cycle time by 40% via MLflow.

Extracurricular Activities

NASA Space Apps Challenge

Oct. 2024

Participant — Seismic Detection Across the Solar System, Egypt

- Developed a seismic signal detection system for planetary data analysis.
- Presented the project in a final demo and discussion session, receiving an achievement certificate.

Technical Skills

Programming Languages: Python, SQL, Java

AI Agents: LangChain, LangGraph, CrewAI, LangSmith, AgentOps, RAG, ReAct, Prompt Engineering

LLMs and NLP: HuggingFace Transformers, Groq, Google Gemini, OpenAI, NLTK, SpaCy

Deep Learning: PyTorch, TensorFlow, Keras, Scikit-learn, XGBoost, OpenCV

Web and APIs: FastAPI, Flask, Streamlit, Gradio

MLOps and Cloud: MLflow, Docker, Kubernetes, GitHub Actions (CI/CD), AWS (EC2, S3, SageMaker), Feast, Prefect

Databases: SQLite, PostgreSQL, SQL Server, Oracle, Pinecone, Chroma, Qdrant

Tools: Git, GitHub, VS Code, Jupyter, Google Colab

Soft Skills: Problem Solving, Critical Thinking, Research & Analysis, Attention to Detail, Fast Learner, Adaptability, Communication, Teamwork & Collaboration

Certifications

- Building LLM Applications using Prompt Engineering — NVIDIA
- Machine Learning Specialization — DeepLearning.AI
- AI Engineer for Data Scientists Associate — DataCamp
- HCIA-AI — Huawei | Python for AI and Data Science — IBM
- Data Scientist — DEPI & EYouth | Applied Deep Learning — MaharaTech
- AI Camp — Sprints × Microsoft

Languages

Arabic: Native

English: Intermediate